

LESSON 5-2 • HOMEWORK PACKET • SHEET 1 OF 2**Area of Triangles**

Maryland Standard 6.G.A.1 • Learning target – I can find the area of a triangle using one half the base times the height.

REMEMBER

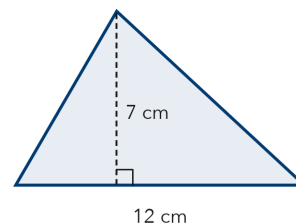
- A triangle is half of a parallelogram with the same base and height: $A = \frac{1}{2} \times b \times h$.
- The height meets the base at a right angle. It can fall **outside** the triangle.
- Area is in **square units** (cm^2 , in^2 , ft^2).

WORKED EXAMPLE

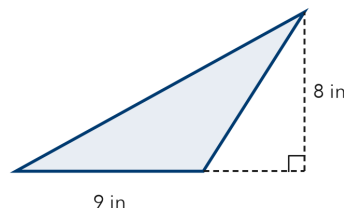
Find the area of a triangle with base 14 cm and height 9 cm.

Step 1 Write the formula: $A = \frac{1}{2} \times b \times h$ **Step 2** Substitute: $A = \frac{1}{2} \times 14 \times 9$ **Step 3** Multiply: $\frac{1}{2} \times 14 = 7$, and $7 \times 9 = 63$ **Answer** 63 cm^2 **PRACTICE***show your work • label units*

1. Find the area of the triangle.

*Show your work.***Answer** _____ cm^2 

2. Find the area of the triangle. The height falls outside the triangle.

*Show your work.***Answer** _____ in^2 

3. Find the area of each triangle with base
- b
- and height
- h
- .

a. $b = 10 \text{ cm}$, $h = 9 \text{ cm}$ _____ cm^2

b. $b = 5 \text{ ft}$, $h = 11 \text{ ft}$ _____ ft^2

- 4.
- Garden Bed.**
- Maya builds a triangular garden bed with a base of 15 feet and a height of 6 feet. One bag of soil covers 9 square feet. How many bags does she need?

*Show your work.***Answer** _____ bags

- 5.
- Find the Error.**
- Jon says the area of a triangle with base 8 m and height 5 m is
- 40 m^2
- . Explain his mistake and find the correct area.

CHECK YOURSELF
☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____
AT HOME Find a triangle at home (a sign, a sandwich half, a roof). Estimate its base and height and its area.

LESSON 5-2 • HOMEWORK PACKET • SHEET 2 OF 2**Missing Measures in Triangles**

Maryland Standard 6.G.A.1 • Learning target – I can find a missing base or height of a triangle when I know its area.

REMEMBER

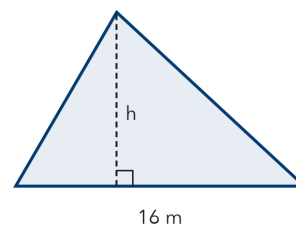
- Know the area and one measure? **Double the area**, then divide by the measure you know: **$h = 2A \div b$** and **$b = 2A \div h$** .
- Check: $\frac{1}{2} \times b \times h$ should equal the given area.

WORKED EXAMPLEArea = 30 in², base = 12 in. Find the height.**Step 1** Double the area: $2 \times 30 = 60$ **Step 2** Divide by the base: $60 \div 12 = 5$ **Step 3** Check: $\frac{1}{2} \times 12 \times 5 = 30$, so it checks**Answer $h = 5$ in****PRACTICE***show your work • label units*

1. Complete the table. Each row describes one triangle.

	Base	Height	Area
Triangle A	8 in		28 in ²
Triangle B		9 ft	54 ft ²

2. The area of the triangle is 48 m
- ²
- . Find the height
- h
- .

*Show your work.***Answer** _____ m

3. Find the missing measure of each triangle.

a. $A = 40$ m², $b = 10$ m; $h =$ _____ m

b. $A = 27$ in², $h = 6$ in; $b =$ _____ in

- 4.
- Sail.**
- A triangular sail has an area of 35 square feet. Its height is 10 feet. How long is its base?

Answer _____ ft

- 5.
- Compare.**
- Triangle P: base 12 cm, height 5 cm. Triangle Q: base 6 cm, height 10 cm. Which has more area? Explain.

TEST PRACTICE*choose the best answer*

6. A triangle has an area of 24 square units and a height of 4 units. What is the length of its base?

A 3 units**B** 6 units**C** 12 units**D** 48 units

7. Which expression gives the area of a triangle with base 18 m and height 5 m?

A 18×5 **B** $\frac{1}{2} \times 18 \times 5$ **C** $18 + 5$ **D** $2 \times 18 \times 5$ **CHECK YOURSELF**
☐ I can do this on my own
 ☐ I need a little help
 ☐ I have a question about #____